ALWAYS A WEIGHTED AVERAGE

Expected value is always computed the same way: each possible value is weighted by its probability and the weighted values are combined. Only the way of combining changes with the type of random variable.

EXPECTED VALUE

$$E[X]$$

split by **the type of random variable**

Σ

CASE 1

discrete

Weighted sum

multiply each value by its probability and add; works for a PMF or for tabulated probabilities

$$E[X] = \sum_i x_i \cdot P(X = x_i)$$

∫

CASE 2

continuous

Weighted integral

multiply each x by its density and integrate; probabilities come from area under the curve

$$E(X) = \int x \cdot f(x)\, dx$$

TAKEAWAY

In both cases the idea is identical: the average of all possible outcomes, each weighted by how likely it is.